

ICLR Research Plan: Budgeted Source Diagnosis for Multimodal Language Models

Detailed execution timeline from June 7 to September 7, with literature positioning, model and dataset choices, baselines, methodology options, risks, and expected results.

Prepared for internal planning. This report treats ICLR 2027 dates as unannounced and uses the ICLR 2026 schedule as a planning proxy. The ICLR 2026 call lists Abstract Submission on Sep 19 and Full Submission on Sep 24, both Anywhere on Earth, and includes uncertainty quantification, representation learning, interpretation, datasets/benchmarks, and safety/ethical considerations within scope [1].

Executive Summary

The ICLR submission should not be a larger benchmark paper or a prompt-routing paper. The strongest trajectory is to formalize multimodal uncertainty as *budgeted source-evidence acquisition*. Given an image, question, model answer, and cheap clean-run signals, a policy decides which diagnostic probe to acquire under a limited forward-pass budget. The final diagnostic state predicts failure family, source-sensitive behavior, and break-aware intervention utility. This makes the paper a learning and representation problem, not an evaluation extension.

The most defensible main claim is: **adaptive source probing recovers most of the diagnostic value of a full multi-probe teacher while using fewer model calls, and it supports break-aware correction decisions without collapsing uncertainty into a single scalar score.**

Recommended Main Direction

The paper should be framed around **Budgeted Source Diagnosis**. The full probe suite is used as a full-evidence teacher during training and analysis. The new contribution is a policy that learns what evidence to acquire, when to stop, and how to support diagnosis under cost. This avoids depending on HalluPrism as a named prior and gives the submission a standalone ICLR identity.

- **Core problem.** Given an MLLM prediction, decide which source probe to run next under a budget.
- **Core method.** Train an adaptive probe policy with stop action, source reconstruction, failure-family prediction, and utility prediction losses.
- **Core result.** Show a cost-diagnostic frontier where adaptive probing beats scalar confidence, random probing, and fixed probe orders at the same forward-pass budget.
- **Core generalization.** Evaluate leave-model-out, held-out dataset, held-out probe severity, and source-shift behavior.
- **Core boundary.** Do not claim causal source decomposition or universal mitigation. Treat routing as break-aware action utility, not as a standalone contribution.

Deep Literature Review And Positioning Gap

The literature review should be written to make one gap unavoidable: recent work either detects hallucination with fixed evidence, decides abstention with scalar or calibrated scores, mitigates hallucination with fixed decoding/intervention rules, or allocates general reasoning compute. It rarely asks which multimodal source evidence should be acquired instance by instance before diagnosis or correction.

Block	Representative Work	What It Gives	Gap For Our Paper
Hallucination benchmarks	POPE, H-POPE, HOPE, HallusionBench, AMBER, IllusionBench, PRE-HAL [2-8]	They expose object, attribute, relation, visual illusion, perception, and reasoning hallucination stresses.	They define stress settings but do not learn a policy for acquiring diagnostic evidence under cost.
Hallucination detection	Evidential conflict, visual-token uncertainty, attention calibration, VaLiD [8-11]	They use features, token uncertainty, attention, or visual-layer behavior to detect or mitigate hallucination.	They usually use a fixed signal. They do not adaptively choose among visual, blank-image, grounding, and relation probes.
Abstention and risk control	Selective prediction and ReCoVERR [12]	They decide when to answer or seek evidence to avoid unnecessary abstention.	They retain answer-versus-abstain as the main interface. Our work adapts the measurement process before diagnosis/correction.

Block	Representative Work	What It Gives	Gap For Our Paper
Adaptive test-time compute	Reasoning-on-a-budget and related TTC surveys [13]	They allocate reasoning or token budget by difficulty.	They decide how much to think. We decide what multimodal evidence to measure.
Active feature acquisition	Active feature acquisition survey [14]	It studies prediction under costly feature measurement.	We instantiate this idea for MLLM reliability, where costly features are behavioral probes rather than tabular fields.

Why This Would Stand Out At ICLR

- **It is a learning problem.** The paper formalizes adaptive probe acquisition, not another fixed hallucination score.
- **It has a clear frontier figure.** Diagnosis quality versus forward-pass cost is a simple ICLR-style result.
- **It links three fields.** The work connects MLLM hallucination, active feature acquisition, and test-time compute under one measurement view.
- **It respects entanglement.** The method does not force independent visual/language/alignment causes. It preserves joint diagnostic structure.
- **It avoids overclaiming mitigation.** Correction is evaluated through utility, fix rate, and break rate, not as a universal routing method.

Base Models To Choose

The main paper should use open models as the primary evidence. This keeps the study reproducible and allows optional logit or hidden-state ablations. Proprietary systems can be appendix-only stress tests if time remains.

Priority	Model	Why Choose It	Role	Risk / Fallback
Must	Qwen2.5-VL-7B-Instruct	Strong open VLM family with visual recognition, object localization, document parsing, dynamic resolution, and long-video features [15].	Primary strong model; leave-model-out anchor.	If memory is tight, use 3B.
Must	LLaVA-OneVision-1.5-8B	Recent open multimodal framework with strong reported benchmark performance and reproducible training recipe [16].	Modern LLaVA-family contrast.	If unavailable, use LLaVA-OneVision-7B or LLaVA-1.6-Mistral.
Must	InternVL3-8B / nearest stable InternVL3 checkpoint	InternVL3 uses native multimodal pretraining and test-time recipes, giving architectural diversity [17].	Second strong non-Qwen family.	If setup is unstable, use InternVL2.5/3.5 small checkpoint.
Must	Molmo-7B-D / Molmo-7B-O	Open-weight and open-data-oriented VLM family; useful for reproducibility and grounding analysis [18].	Open-data contrast model.	If runtime is high, use as held-out model only.
Stretch	PaliGemma 2-10B/28B or Gemma 3/4 vision model	PaliGemma 2 provides a transfer-focused VLM family across sizes/resolutions [19].	Transfer and model-family robustness.	Use only if compute remains healthy.

Minimum viable model set: Qwen2.5-VL-7B, LLaVA-OneVision-8B, InternVL3-8B, and Molmo-7B. Use PaliGemma/Gemma as a stretch model or held-out architecture.

Datasets To Choose

The dataset suite should be source-stress-driven. The goal is not to create another general VQA leaderboard. The suite should create meaningful pressure on object priors, visual fragility, visual illusion, relation binding, and answerability.

Type	Dataset	Diagnostic Stress	Role In Paper	Sampling Plan
Core	POPE plus HOPE or H-POPE	Object hallucination and object-prior pressure [2-4].	Language-prior and object hallucination stress.	Use balanced subset, 3K-5K per model if full POPE is too costly.
Core	HallusionBench	Entangled language hallucination and visual illusion [5].	Mixed failure-family diagnosis and ambiguity.	Use image-paired subset.
Core	VizWiz-VQA	Visual insufficiency, low-quality images, answerability.	Visual fragility and answerability stress.	Use validation split or balanced subset.
Core	VSR plus small GQA relation slice	Spatial and relation binding.	Relation probe and alignment stress.	Use all VSR; add GQA slice if relation coverage is too small.
Held-out	IllusionBench	Classical and real-world visual illusions [6].	Visual illusion transfer stress.	Use held-out only, 1K-2K QA pairs.

Type	Dataset	Diagnostic Stress	Role In Paper	Sampling Plan
Held-out	PRE-HAL	Perception and reasoning hallucination across instances, scenes, and relations [8].	Perception/reasoning source shift.	Use held-out only; relation subset is valuable.
Appendix	AMBER or MMHal-Bench	Existence, attribute, and relation hallucinations [7].	External hallucination sanity check.	Use if pipeline is stable.

Baselines To Include

The baseline set must make adaptivity necessary. A strong paper should not only show that probes help; it should show that a learned policy chooses probes better than fixed or random acquisition at equal cost.

Class	Baseline	Purpose	Required?
Scalar	Scalar confidence, entropy, margin	Tests whether ordinary confidence ranks enough diagnostic risk.	Must
Clean-only learned	Logistic regression / MLP from clean features	Tests whether probing is needed after observing clean output.	Must
Fixed acquisition	Visual-first, blank-first, grounding-first, relation-first	Tests whether a simple fixed order is enough.	Must
Random acquisition	Random probe order under same budget	Controls for added compute alone.	Must
Full evidence	All probes, full teacher	Upper bound for diagnostic evidence.	Must
Uncertainty-greedy	Next probe chosen by predicted entropy or confidence gap	Simple adaptive heuristic.	Must
Literature-aligned	ReCoVERR-style evidence collection	Connects to selective VLM answering.	If feasible
Detection-oriented	Evidential conflict / visual-token uncertainty / VaLiD-style signals	Compares against recent detection and mitigation lines.	Appendix if feasible
Action	No correction, generic correction, fixed source correction, learned action selection	Tests break-aware decision support.	Must for utility section

Methodological Directions

The primary method should be an adaptive probe policy. Secondary methods should support ablations, not compete for the main story. The final paper should not become a bag of variants.

Direction	Technical Design	Why It Stands Out	Expected Evidence
Full-evidence teacher	Run clean prediction, visual probes, blank-image probe, grounding probe, relation probe when available, and correction branches.	Creates behavioral supervision without needing human labels for every instance.	Teacher upper bound and source-evidence labels.
Adaptive probe policy	Sequential actions: STOP, VISUAL, BLANK, GROUND, RELATION. State updates from answer flip, confidence shift, match, and probe type.	Transforms uncertainty into active measurement under cost.	Cost-diagnostic frontier dominates fixed/random policies.
Entanglement-preserving representation	Joint source state with source reconstruction and failure diagnosis heads; no orthogonality constraint.	Matches empirical fact that source signals overlap.	Ablation shows forced independence weakens diagnosis.
Budget-conditioned inference	At budgets B=1,2,3,4,full, choose probes and stop when evidence is enough.	Directly connects to ICLR test-time compute interest.	Smooth frontier, useful low-budget operating points.
Break-aware utility	Utility = fix - beta * break - gamma * cost. Action utility predicted from diagnostic state.	Turns routing into decision theory, not prompt engineering.	Lower regret or break rate than generic correction in high-risk slices.
White-box optional	Add logits, hidden norms, visual-token statistics where available.	Shows whether black-box probes remain competitive.	Appendix table; not necessary for main claim.

Detailed Weekly Timeline: June 7 To September 7

This schedule assumes one PhD student as the primary executor, with weekly faculty review and one mid-week engineering check. The plan is intentionally front-loaded on literature and pilot evidence. By June 30, the team should know whether the work is truly adaptive evidence acquisition rather than a cheaper fixed-probe variant.

Week	Dates	Main Focus	Deliverables	Gate / Decision
1	Jun 7-13	Literature review and identity freeze.	Five-block literature matrix; 2-page positioning memo; final title family; novelty paragraph; baseline list derived from literature.	Paper must be explainable without naming prior internal work.
2	Jun 14-20	Experimental scope freeze.	Model availability check; dataset access check; sample counts; compute budget spreadsheet; exact prompt families.	Freeze 4 core models and 4 core datasets. Remove any unstable model early.
3	Jun 21-27	Clean inference harness.	Loaders; clean prompts; answer normalization; confidence extraction; 50-example smoke test per dataset per model.	Two models must run all datasets end-to-end with valid logging.
4	Jun 28-Jul 4	Full-evidence teacher pilot.	Visual, blank, grounding, relation probes; teacher records on 500 examples; probe sanity plots.	Teacher signal must show nontrivial variation. If saturated, revise probes.
5	Jul 5-11	Formal method and split design.	BSD formulation; actions and stop rule; loss design; train/dev/test splits; leave-model-out and held-out dataset split files.	Method must read as active feature acquisition for MLLMs.
6	Jul 12-18	Scale teacher data for first two models.	Qwen2.5-VL and LLaVA-OneVision full teacher records on core datasets; start IllusionBench held-out sample.	Generation failure below 2-3%. Normalization audit acceptable.
7	Jul 19-25	Non-adaptive baselines.	Scalar, clean-only learned, fixed probe orders, random acquisition, uncertainty-greedy, full teacher upper bound.	At least one evidence baseline beats scalar confidence for diagnosis.
8	Jul 26-Aug 1	Adaptive policy v1.	Sequential policy with STOP; source reconstruction head; failure-family head; first cost-diagnostic frontier.	Adaptive policy beats random and at least two fixed orders at equal budget.
9	Aug 2-8	Complete minimum model set.	Molmo and InternVL teacher data; retrained policy; 4-model in-distribution results.	Keep PaliGemma/Gemma only if it does not delay main run.
10	Aug 9-15	Generalization tests.	Leave-model-out; held-out IllusionBench; PRE-HAL subset if feasible; probe-severity shift.	Leave-model-out must be directionally positive. If held-out transfer is weak, report as stress test.
11	Aug 16-22	Break-aware utility.	Utility labels; no/generic/fixed-source/adaptive action comparisons; fix, break, accuracy, utility, regret.	Frame as decision support. Do not claim universal mitigation.
12	Aug 23-29	Ablations and reliability.	No source loss; no stop; no flip; no confidence shift; no relation; forced-independent vs joint; bootstrap CIs.	Ablations must support adaptivity and source evidence as necessary.
13	Aug 30-Sep 5	Full draft and figure lock.	Complete first draft; Figure 1 pipeline; Figure 2 frontier; model/dataset table; main result tables; appendix skeleton.	No new main experiments after Sep 5 unless there is a fatal error.
14	Sep 6-7	Internal submission freeze.	Executive summary; final experiment freeze; reviewer-risk memo; checklist of missing citations and licenses.	Ready for full polishing cycle after Sep 7.

Expected Headline Results

- **Cost frontier.** Adaptive probing reaches 70-90% of full-teacher diagnostic gain using at most 3-4 forward passes on average.
- **Model generalization.** Leave-model-out adaptive probing outperforms scalar confidence, clean-only prediction, and fixed probe orders.
- **Dataset stress.** Held-out IllusionBench or PRE-HAL shows partial transfer and identifies source-shift failure modes honestly.
- **Decision utility.** Break-aware action selection lowers breakage or regret compared with generic correction in high-risk slices.

- **Ablation clarity.** Removing the stop action, source reconstruction, or probe feedback weakens the frontier. This supports the mechanism.

Figures And Tables Needed For A Wow Submission

Item	Content	Purpose
Figure 1	Adaptive source-evidence acquisition pipeline: clean prediction, policy action, probe output, diagnostic state, final diagnosis/action.	Makes the method visually obvious.
Figure 2	Diagnosis AUROC/Macro-F1 versus average forward passes.	Main ICLR frontier figure.
Figure 3	Probe acquisition heatmap by dataset/source stress.	Shows learned behavior is interpretable.
Table 1	Models, datasets, source stresses, sample counts, probe availability.	Scope and reproducibility.
Table 2	Budgeted diagnosis across baselines.	Main quantitative result.
Table 3	Leave-model-out and held-out dataset results.	Generalization evidence.
Table 4	Break-aware utility, fix, break, accuracy, regret.	Actionability without overclaiming routing.
Table 5	Ablations.	Mechanism evidence.

Evaluation Metrics

- **Diagnostic quality:** weighted AUROC, Macro-F1, balanced accuracy, per-source/failure-family Macro-F1.
- **Source reconstruction:** MAE for source scores, Spearman correlation, top-source agreement, calibration of predicted source risk.
- **Cost:** average forward passes, median forward passes, cost-normalized AUROC, area under cost-diagnostic frontier.
- **Decision utility:** accuracy, fix rate, break rate, expected utility, regret against full-teacher oracle.
- **Generalization:** leave-model-out, leave-dataset-out, held-out severity, held-out source stress.
- **Reliability:** bootstrap CIs, paired tests for frontier area, and slice-level error analysis.

Risk Mitigation Plan

Risk	Likely Symptom	Mitigation
Teacher saturation	One source score dominates all examples.	Use teacher as joint evidence, add match-only/stable controls, emphasize full vector and normalized policy state.
Weak adaptive gain	Fixed probe order matches learned policy.	Improve state features, train budget-conditioned selector, add stop-action calibration, or make fixed-order strength a transparent boundary.
Poor held-out transfer	Held-out dataset results weak.	Use leave-model-out as main generalization; report held-out data as source-shift stress; add domain-adversarial or group-balanced training if time permits.
Compute overrun	Full teacher runs miss schedule.	Downsample POPE/HOPE, drop stretch model, keep four core models, reserve held-out stress for smaller subsets.
Routing overclaim	Reviewers see prompt engineering.	Use break-aware utility and regret. Do not call routing a main contribution.
Self-plagiarism concern	New work looks like fixed-probe paper.	Frame full probes as teacher; make adaptive acquisition and cost frontier the main method and result.

Minimum Viable Submission By September 7

By September 7, the project should have a full draft and frozen main results, not merely a plan. The minimum viable submission needs four open models, four core datasets, one held-out stress set, scalar/fixed/random/clean/full-teacher baselines, one adaptive policy, leave-model-out results, and one break-aware utility analysis. The stretch version adds PRE-HAL, PaliGemma/Gemma, white-box features, and additional mitigation baselines.

Recommended Paper Claim

We formalize multimodal uncertainty as adaptive source-evidence acquisition. A learned probing policy decides which visual, blank-image, grounding, or relation evidence to acquire under budget. Across open MLLMs and source-stress benchmarks, adaptive probing improves the diagnosis-cost frontier over scalar confidence,

clean-only prediction, random probing, and fixed probe orders. The resulting diagnostic state supports break-aware correction decisions without reducing source structure to a scalar abstention score.

References

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